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Towards the Evaluation of Similarity in Representations Embedded by CEBRA

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Abstract CEBRA extends nonlinear independent component analysis to generate consistent representations of latent structures from high-dimensional data. While it has demonstrated reconstruction performance and linear identifiability, validation of CEBRA embeddings remains challenging. In this study, CEBRA is employed to compute the embedding of each multimodal observation generated from the common latent factors. The common latent space is subsequently computed by aligning the obtained embeddings using canonical correlation analysis. The results show that this approach effectively captures high correlations between observations, overcoming potential confounds from high-dimensional randomness.

1 Introduction

The advent of deep learning has led to a paradigm shift in preprocessing, replacing traditional manual methods with innovative approaches that use neural networks to generate effective representations. This process is referred to as representation learning. It constitutes a critical component of contemporary machine learning methodologies. Representations learned by deep neural networks are widely used in many machine learning applications, such as speech recognition and natural language processing.

In the field of representation learning, identifiability has been primarily studied in the context of Independent Component Analysis (ICA). Due to the lack of identifiability or uniqueness of the representation, extending ICA to the nonlinear case was problematic. Recently, nonlinear extensions using time structure or some auxiliary information have been proposed. The existence of such models has been well studied, resulting in an increasing proliferation of algorithms dedicated to their identifiability [1].

An example of the identifiability of a nonlinear ICA with auxiliary variables is CEBRA [2]. Using a generalized contrastive learning algorithm, CEBRA introduced a new paradigm for joint modeling of time-series data [3]. The data sampling scheme and model optimization are directly linked. The resulting latent factors can be interpreted based on input auxiliary variables (e.g. behavior, animal identification, reward, estimated internal state). CEBRA has been shown to significantly outperform other algorithms in generating consistent embeddings and producing visually informative embeddings.

By applying CEBRA to each of the multimodal observations generated from common latent factors, it can be expected to obtain an embedding extracted at a linearly identifiable level for each. Due to this linear identifiability, these embeddings are expected to be aligned using linear Canonical Correlation Analysis (CCA). On the basis of this assumption, we propose a method to identify a common latent structure using CEBRA and CCA.

2 Related Works

2.1 CEBRA

Using user-defined auxiliaries (hypothesis-driven) or time-only labels (discovery-driven), CEBRA obtains a consistent embedding of neural activity that can be used for downstream tasks such as data visualization and decoding. Specifically, CEBRA instantiates a nonlinear ICA based on contrastive learning.

Contrastive learning is a method that uses contrasting positive and negative samples to find attributes that they have in common, as well as attributes that distinguish them. Generalized contrastive learning [4], as introduced in CEBRA, builds on the core idea of traditional contrastive methods by introducing a more flexible framework for representation learning. Instead of focusing solely on a fixed pair of positive and negative examples, it employs a randomized binary identification task that can incorporate multiple forms of relationships within the data. This flexibility enables the model to capture a broader spectrum of intrinsic features and complex relationships, resulting in richer and more robust representations across various tasks and modalities.

Figure 1 illustrates the CEBRA paradigm. It is possible to learn a mapping from the observable data K to the potential dynamics Z.

2.2 Canonical Correlation Analysis

Canonical Correlation Analysis (CCA) is a statistical method that identifies linear combinations of two sets of variables in such a way that the correlation between these combinations is maximized [5]. CCA is well suited for a wide range of applications, including neuroscience, because it can uncover common sources of statistical variation across multiple modalities without imposing predefined directional assumptions [6].

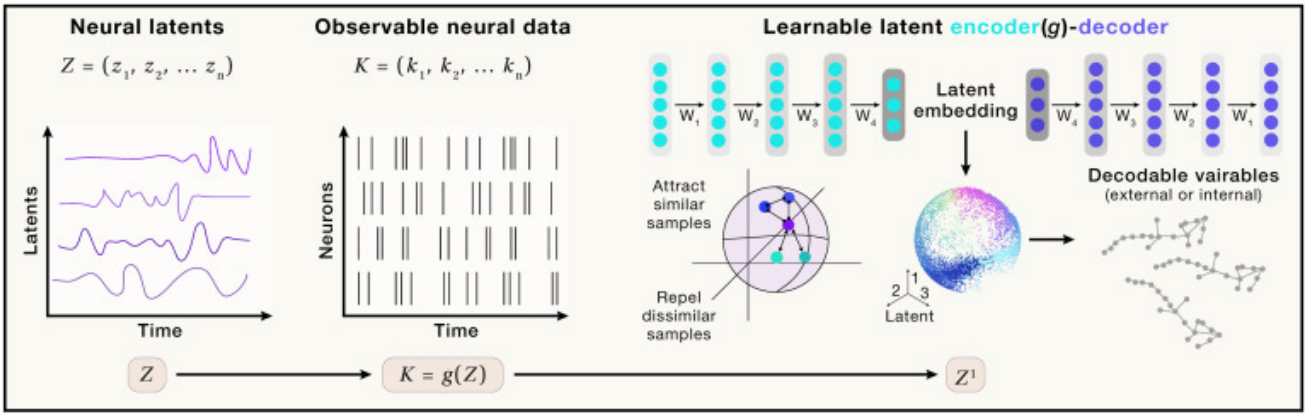


Figure 1: Process of latent structure extraction using CEBRA. Cited from Fig.4 in [3]

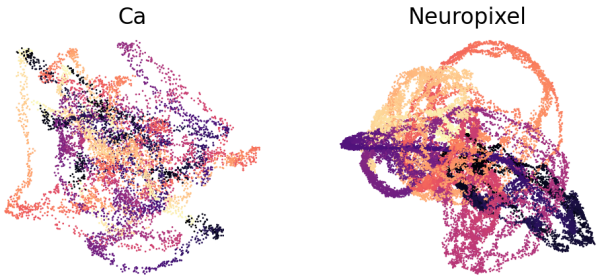


Figure 2: Visualization of trained 128 dimensional latent CEBRA embeddings (without auxiliary labels)

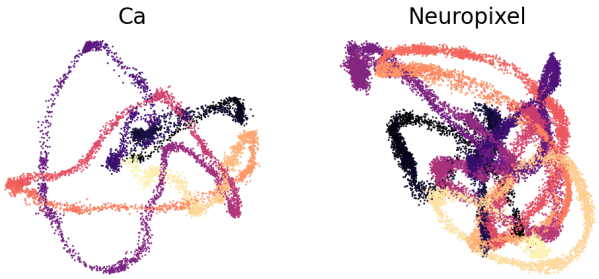


Figure 3: Visualization of trained 128 dimensional latent CEBRA embeddings (with auxiliary labels)

3 Evaluation of Similarity between Embeddings Using CCA

For each multimodal observation generated from common latent factors, CEBRA is applied to compute the embedding. CEBRA provides a latent structure extracted at a linearly identifiable level. The common latent space is computed by aligning the embeddings obtained by means of the CCA. We expect the latent space obtained by the CCA to represent the true common latent structure.

4 Experiments

We evaluated our proposed method using the Allen Brain Observatory dataset [7][8]. In this dataset, neural data from Neuropixel (Np) probes or two-photon microscopy calcium imaging (Ca) were recorded as mice passively watched a video (30 seconds, 30 Hz) repeatedly. The features of the video were extracted frame by frame using DINO [9].

4.1 CEBRA Embeddings

Embeddings were computed from neural data from mice during video viewing using CEBRA. The Ca and Np embeddings calculated by CEBRA without auxiliary (time-only labels) are shown in Figure 2. The Ca and Np embeddings calculated by CEBRA with auxiliary (movie labels) are shown in Figure 3. As an auxiliary information for CEBRA, the 768 dimensional features[1] extracted from the video frames using DINO were used.

4.2 Alignment of Embeddings Using CCA

The similarity between Ca, Np, and Movie (the DINO features of the video) was calculated using CCA. The number of canonical components to be computed was set to 10. The CCA was calculated using mvlearn [10], a Python library that implements typical multi-view machine learning methods, including multi-view CCA [11] for two or more views in addition to basic CCA.

5 Results

In this paper, we report results with auxiliary labels. Table 1 shows the CCA results for Ca (Rank: 128) and Movie (Rank: 662). There was a high correlation between Ca and Movie. The top three components of the two views are shown in Figure 4. Similarly, Tables 2 and 3 show the CCA results for Np (rank: 128) and Movie, the CCA results for Ca and Np, respectively. The top three components of the two

[1]The rank is 662.

Table 1: CCA between Movie and Ca with auxiliary labels

	r	Wilks	p-val
1-10	0.999	9.395e-22	1.110e-16
2-10	0.998	3.914e-19	1.110e-16
3-10	0.998	1.297e-16	1.110e-16
4-10	0.997	2.613e-14	1.110e-16
5-10	0.996	4.564e-12	1.110e-16
6-10	0.995	5.877e-10	1.110e-16
7-10	0.994	5.597e-08	1.110e-16
8-10	0.993	4.921e-06	9.084e-04
9-10	0.992	3.719e-04	1.000
10-10	0.989	2.234e-02	1.000

Table 2: CCA between Movie and Np with auxiliary labels

	r	Wilks	p-val
1-10	0.999	7.587e-22	1.110e-16
2-10	0.998	3.767e-19	1.110e-16
3-10	0.998	9.910e-17	1.110e-16
4-10	0.997	2.189e-14	1.110e-16
5-10	0.996	3.617e-12	1.110e-16
6-10	0.995	4.810e-10	1.110e-16
7-10	0.994	4.724e-08	1.110e-16
8-10	0.993	3.967e-06	2.698e-12
9-10	0.992	2.943e-04	1.000
10-10	0.990	1.912e-02	1.000

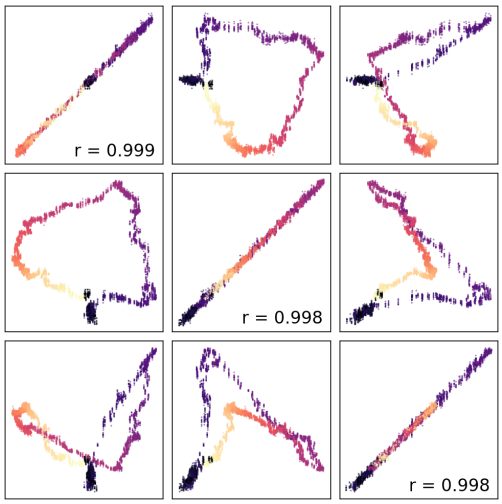


Figure 4: Common latent structure between Ca and Movie with auxiliary labels

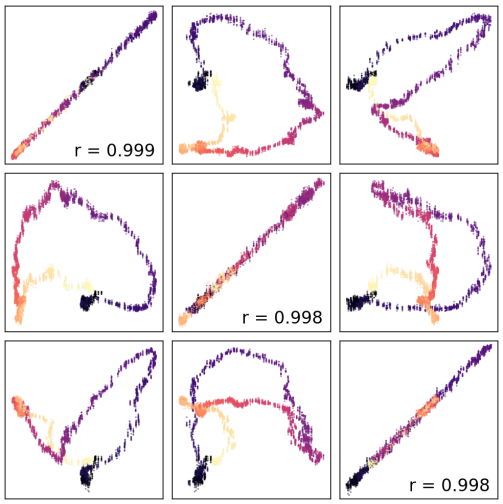


Figure 5: Common latent structure between Np and Movie with auxiliary labels

views are shown in Figures 5 and 6, respectively. Despite the complex shape embedding in Figures 2-3, the individual plots in the common latent space in Figures 4-6 show a simple shape. The observation of a simple shape in the aligned latent space may suggest that the high-dimensional embeddings have effectively converged onto a common latent structure. This qualitative indication is supported by the quantitatively high correlation coefficients observed in the analysis.

6 Discussion

The correlations between Ca and Movie, Np and Movie, and Ca and Np were shown to be higher than 0.99[2]. This result suggests that latent structures extracted from multimodal observations generated from common latent factors have a common latent structure.

Thus, our approach demonstrates promise by leveraging the inherent linear identifiability of embeddings generated by CEBRA, which allows for the effective use of CCA to align multimodal observations. The high correlation coefficients observed between the modalities provide empirical support for the underlying hypothesis that a common latent structure can be extracted through linear alignment, thus validating the method under the given assumptions.

Nevertheless, several limitations warrant careful consideration. First, reliance on linearity may not adequately capture the complex, nonlinear relationships often present in real-world data, potentially limiting the applicability of the method in scenarios where the underlying structure deviates from linearity. Second, the evaluation was confined to a single dataset (i.e., the Allen Brain Observatory), raising concerns about the generalizability of the results across different data domains and conditions. Finally, the exclusive focus on

[2]Note that the correlation coefficient was markedly lower when the latent structure obtained by CEBRA was shuffled (See Appendix B).

Table 3: CCA between Ca and Np with auxiliary labels

	r	Wilks	p-val
1-10	0.998	6.313e-22	1.110e-16
2-10	0.998	2.059e-19	1.110e-16
3-10	0.997	4.933e-17	1.110e-16
4-10	0.997	9.491e-15	1.110e-16
5-10	0.996	1.554e-12	1.110e-16
6-10	0.996	1.968e-10	1.110e-16
7-10	0.995	2.351e-08	1.110e-16
8-10	0.995	2.499e-06	1.110e-16
9-10	0.993	2.400e-04	1.110e-16
10-10	0.991	1.801e-02	1.110e-16

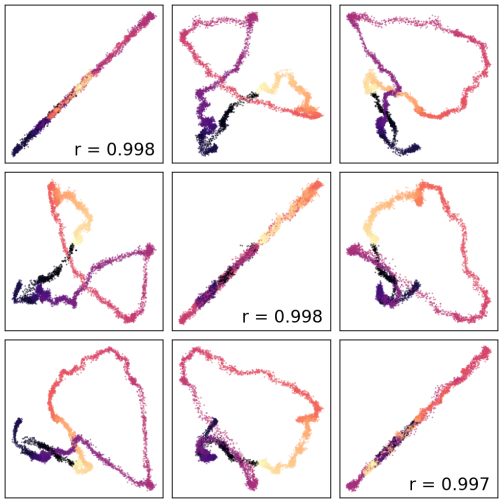


Figure 6: Common latent structure between Ca and Np with auxiliary labels

linear CCA without comparison to alternative alignment techniques such as nonlinear manifold alignment methods, deep learning-based approaches, or Procrustes analysis restricts the context in which the method's performance and limitations can be fully understood. Future research should incorporate a broader range of alignment strategies to more comprehensively assess the strengths and weaknesses of the proposed approach.

7 Conclusion

In this study, we proposed a novel method that leverages the linear identifiability of CEBRA embeddings and aligns them using linear CCA to uncover a common latent structure across multimodal observations. Our experiments demonstrated that this approach captures high correlations between the observations, overcoming potential confounds due to high-dimensional randomness. These findings not only validate our method, but also open new avenues for integrating multimodal data in predictive models, which will be explored in future work.

Acknowledgments

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A Model Parameters and Settings for CEBRA

In this appendix, we provide detailed information regarding the hyperparameter settings used in our experiments with CEBRA. The following settings were employed:

```
time_offsets: 10,
distance: 'cosine',
output_dimension: 128,
batch_size: 512.
```

For the Allen datasets, different model architectures were utilized based on the modality:

```
For Allen datasets (Ca),
model_architecture: 'offset10-model'

For Allen datasets (Np),
model_architecture: 'offset40-model-4x-subsample'
```

B Effects of Feature Shuffling on Embedding Alignment

To ensure that the high correlations observed between the CEBRA embeddings are not merely artifacts of high-dimensional data randomness, we performed an additional analysis in which the extracted features were randomly shuffled. The shuffling procedure disrupts any inherent structural relationships present in the data. Figure 7 reveals that, upon shuffling, the correlation coefficients drop markedly. which confirms that the alignment achieved through linear CCA is attributable to the underlying latent structure rather than chance. This analysis underscores the importance of linear identifiability in reliably extracting shared latent factors from multimodal observations.

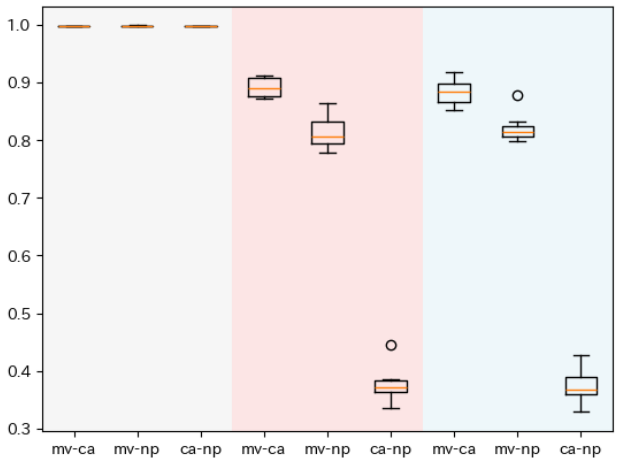


Figure 7: Comparison of CCA scores under different shuffling conditions. The yellow background represents the results without any shuffling, the red background corresponds to shuffling a set of features, and the blue background indicates shuffling each feature independently. These comparisons demonstrate that random shuffling significantly reduces the correlation coefficients.